CUSTrack: Causality-Inspired Liver Ultrasound Tracking With Periodic Motion Bias Mitigation

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***Abstract*—Real-time tissue tracking is a fundamental task in liver ultrasound applications. Due to the periodic nature of liver motion, historical trajectories can offer valuable priors for target localization, particularly when foreground–background distinction is weak. However, existing trackers often exploit these trajectories as shortcuts, relying excessively on periodic respiratory patterns rather than true object appearance matching. In this work, we revisit liver tracking from a causal perspective and propose CUSTrack, a method that mitigates periodicity bias by decomposing and correcting the total causal effect of historical trajectories. We define *periodicity bias* as the *direct causal effect* of past states and eliminate it via counterfactual reasoning, preserving 'good' trajectory priors while suppressing 'bad' periodic bias. To ensure identifiability, we incorporate a deconfounding module that removes latent confounders from fused feature representations. Extensive experiments on liver ultrasound datasets demonstrate that CUSTrack achieves superior tracking accuracy and robustness under challenging conditions.**

***Index Terms*—Liver ultrasound tracking, periodicity bias, counterfactual reasoning.**

I. INTRODUCTION

REAL-TIME tracking plays a critical role in ultrasound-guided interventions such as minimally invasive surgery [1] and radiotherapy [2]. In particular, accurate localization of anatomical structures is essential for the precise delivery of radiation doses to target tissues. Given an initial target position annotated in the first frame, the tracking system aims to continuously estimate the location of the target across successive ultrasound frames.

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From the perspective of visual tracking, template matching remains the dominant paradigm in existing studies [3], [4]. These methods typically adopt shared feature extraction backbones to focus on *how to describe* the target, followed by a matching module that determines *where to locate* it. Whether using cross-correlation [3] or attention-based fusion [5], the location with the highest similarity is selected as the predicted target. However, in practical ultrasound imaging, these models often suffer from degraded performance due to challenges such as low signal-to-noise ratio, occluded or incomplete anatomical structures, and the presence of distractors with similar appearance. These difficulties are further exacerbated by variations in landmark depth, size, and morphology. To address these limitations, recent efforts have explored robust feature representations (e.g., self-supervised learning [6]) and template updating strategies [7] that dynamically adapt to target deformation. Nevertheless, template matching alone remains insufficient when the target structure becomes partially invisible or temporarily lost, such as when anatomical landmarks move out of the imaging plane.

Inspired by the human perception of moving objects, another stream of research emphasizes the use of preceding trajectories. A straightforward approach involves treating previous target positions as spatial priors to suppress implausible matching regions and maintain temporal consistency throughout the tracking process [8]. Although incorporating historical trajectories can enhance the robustness of the model—particularly when the target structure is partially missing or temporarily obscured—it also introduces a critical challenge: **periodicity bias**.

Consider a tracker model that leverages historical states, such models tend to focus on those most readily learnable patterns to optimize their objective loss, particularly exploiting spurious correlations between respiratory phases and tissue positions [9]. As illustrated in Fig. 1, respiratory-induced liver motion characteristically exhibits periodic behavior, cycling repeatedly from end-expiration to end-inspiration phases [10]. Periodicity bias manifests when trajectory predictions are overly influenced by these cyclic patterns rather than the true

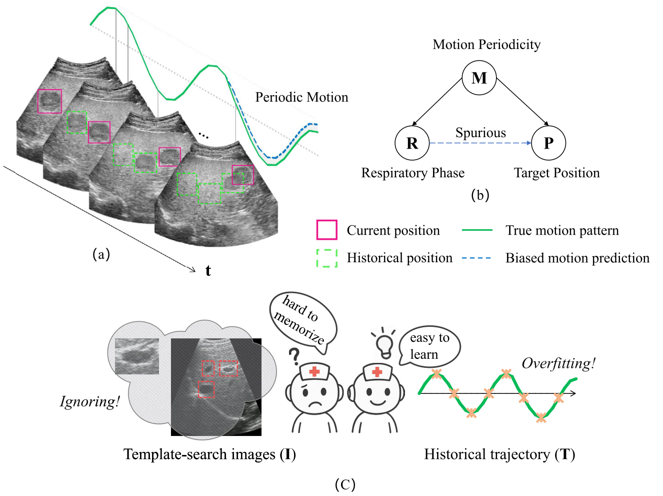


Fig. 1. (a) Periodic liver motion: Respiratory-induced liver motion typically follows repetitive cycles from end-expiration to end-inspiration; (b) Periodicity bias: From a causal point, trackers with historical liver positions may infer the liver position utilizing spurious associations between the respiratory phase (R) and target position (P), denoted by blue dotted line, forming biased predictions; (c) From a learning point, these model overfit periodic motion priors and neglect visual features, making predictions solely by historical trajectories.

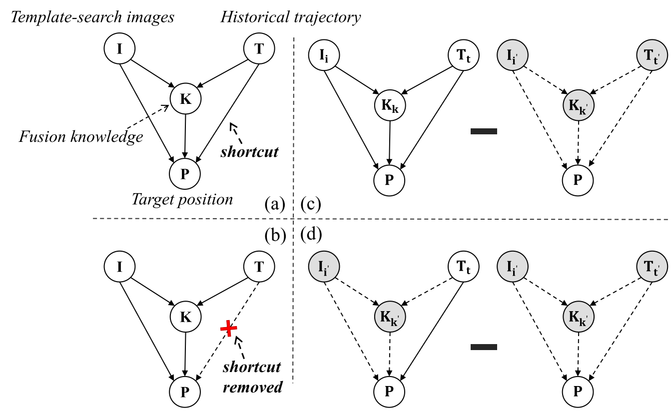


Fig. 2. Causal graph and path-specific effect illustration. (a) Original structure with T $\rightarrow$ P; (b) Adjusted structure with the direct trajectory path removed; (c) Total effect (TE); (d) Natural direct effect of trajectory (NDE$_T$).

target structure. This phenomenon is especially intractable when landmark motion deviates from expected periodic patterns due to factors such as abrupt patient movements or hand-held probe instability [11]. For these case, periodicity bias can significantly compromise prediction accuracy of the tracking model.

While existing approaches attempt to balance visual and temporal information, they lack a principled framework to distinguish beneficial motion priors from harmful periodic dependencies. Understanding and addressing these periodic dependencies is therefore critical to advancing ultrasound tracking systems. In this study, we revisit **periodicity bias** from a causal perspective. As illustrated in Fig. 1, respiratory-induced liver motion introduces periodicity as a latent confounder, forming shortcut associations between respiratory phase and target position. We conceptualize trajectory predictions as the joint outcome of two contributing factors: 1) intrinsic states of the target structure, including its position, shape, and temporal dynamics; and 2) periodic patterns induced by respiration. To effectively leverage motion periodicity, we disentangle the effect of historical trajectories into a '*good*' component—location priors that constrain the search space and exclude background—and a '*bad*' component—periodicity bias that over-associates target position with respiratory phase. This work aims to retain the benefits of location priors while mitigate the adverse effects of periodicity bias in ultrasound tracking.

In this paper, we present **CUSTrack**, a causality-inspired liver ultrasound tracking framework designed to mitigate periodicity bias induced by cyclic organ motion. As illustrated in the causal graph in Fig. 2, CUSTrack provides a principled approach to alleviate periodicity bias. In a conventional tracker, the target position (P) is estimated as the joint effect of template-search image input (I) and historical trajectory information (T). However, such trackers often conflate the *beneficial* location priors with the *harmful* over-reliance on periodic patterns, resulting in degraded robustness when landmarks deviate from expected cycles.

To disentangle these effects, CUSTrack employs a principled causal framework with three key components. First, a **counterfactual tracker** isolates the direct trajectory influence ($T \rightarrow P$) by estimating predictions based solely on historical motion states. Second, a **confounder correction module** ensures valid causal inference by disentangling latent factors affecting both visual and motion features. Finally, the framework generates **calibrated predictions** by removing the isolated periodicity bias from total predictions, emphasizing image-driven analysis while preserving beneficial location priors. The key contributions are summarized as follows:

- We formalize periodicity bias in ultrasound tracking from a causal inference perspective, providing theoretical foundations for understanding trajectory-induced shortcuts.
- We develop CUSTrack, a causality-inspired framework that systematically disentangles beneficial location priors from harmful periodicity bias through counterfactual reasoning.
- We introduce a confounder correction module that ensures identifiable causal effect estimation, enabling robust fusion of visual and temporal information.

II. Related Work

A. Trackers Without Historical States

Conventional visual tracking methods rely on appearance matching without explicitly modeling temporal dynamics. Siamese-based trackers [3], [4] correlate templates with candidates, while Transformer-based models [12], [13] enhance feature interaction via attention. Recent sequence models such as SeqTrack [14] adopt autoregressive decoding but still lack explicit motion history modeling. In ultrasound tracking, efforts like self-supervised learning [6], dynamic template updating [15], and multi-scale fusion [16] mitigate appearance variation, yet largely ignore temporal continuity

TABLE I
NOTATIONS AND DESCRIPTIONS

Notation	Description
I	Template-search image pair (visual input).
T	Historical trajectory states (temporal input).
K	Fusion representation derived from I and T.
$\tilde{K}$	Deconfounded fusion representation with latent bias removed.
P	Target position.
U	Latent confounders affecting I, T, or both.
I_i, T_t, K_k	Variable assignments (e.g., $I_i \triangleq I = i$).
$P(I_i, T_t, K_k)$	Predicted position under assignments I_i, T_t, K_k.
$\mathcal{A}_i, \mathcal{A}_t, \mathcal{A}_k$	Contribution of $i/t/k$ input to the prediction (effect components).
$\mathcal{P}_{i,t,k}$	Predicted position under specific input configuration (i, t, k).
$E[\cdot]$	Expectation operator over the outcome under stochastic data distribution, approximated by the empirical mean of model predictions.

and remain vulnerable to respiratory-induced periodic motion, which introduces systematic biases unique to medical imaging.

B. Trackers With Historical States

Temporal modeling has shown promise in enhancing tracking robustness. Methods such as STMTrack [17] and ARTrack [18] leverage memory networks or autoregressive dynamics to incorporate trajectory priors. However, most approaches [18], [19], [20] treat trajectory merely as auxiliary input, without distinguishing between informative spatial constraints and harmful periodic shortcuts. This limitation is particularly critical in medical imaging, where periodic respiratory motion induces spurious temporal correlations [10], motivating our causal framework that disentangles beneficial location priors from periodicity bias via counterfactual reasoning.

C. Causality-Inspired Vision

Causal reasoning offers a principled way to disentangle true causal effects from spurious correlations, and has been applied to vision tasks such as long-tail recognition [21], domain generalization [22], and multimodal understanding [23], [24]. However, existing efforts largely focus on static visual or semantic biases, while the temporal confounding in sequential tracking—where historical information serves as both helpful priors and harmful shortcuts—remains underexplored. In medical imaging, causal perspectives have been introduced to handle confounders and improve generalization [25], [26], [27]. This challenge is particularly salient in medical imaging, where periodic organ motion induces systematic biases. We extend causal frameworks to this setting by mitigating trajectory-induced periodicity bias in ultrasound tracking.

III. PRELIMINARIES

A. Notations

The key notations used in this paper are summarized in Table I.

B. Causal Graph and Causal Effects

We formulate the causal structure of ultrasound tracking as a directed acyclic graph (DAG) over variables $\{I, T, K, P\}$, as illustrated in Fig. 2(a). Both I and T affect P directly ($I \rightarrow P$, $T \rightarrow P$) and indirectly via fusion representation K ($I \rightarrow K \rightarrow P$, $T \rightarrow K \rightarrow P$). The stochastic fusion of I and T into K introduces a probabilistic mediation pathway.

To quantify how each pathway contributes to predictions, we decompose the causal influence into three key effects:

1) Total Effect (TE): TE captures the overall impact on prediction P when changing both inputs (Fig. 2(c)). Formally:

$$\mathrm{TE} = E[P(I_i, T_t, K_k)] - E[P(I_{i'}, T_{t'}, K_{k'})] \tag{1}$$

where (I_i, T_t) and $(I_{i'}, T_{t'})$ are two input configurations, and K_k, $K_{k'}$ are the corresponding fused representations associated with these inputs. $E[\cdot]$ denotes the expectation over the outcome under stochastic distributions.

2) Natural Direct Effect of Trajectory (NDE_T): NDE_T quantifies the trajectory's isolated influence through the direct path $T \rightarrow P$. To achieve this isolation, we use a counterfactual approach:

$$\mathrm{NDE}_T = E[P(I_{i'}, T_t, K_{k'})] - E[P(I_{i'}, T_{t'}, K_{k'})] \tag{2}$$

Here, image input $I_{i'}$ and fusion knowledge $K_{k'}$ remain fixed while trajectory varies from $T_{t'}$ to T_t. This blocks the mediated path $T \rightarrow K \rightarrow P$, allowing us to isolate the direct influence of trajectory on prediction, as illustrated in Fig. 2(d).

3) Adjusted Causal Effect (ACE): ACE captures the remaining influence after removing trajectory's direct effect:

$$\begin{aligned} \mathrm{ACE} &= \mathrm{TE} - \mathrm{NDE}_T \\ &= E[P(I_i, T_t, K_k)] - E[P(I_{i'}, T_t, K_{k'})] \end{aligned} \tag{3}$$

Both terms share trajectory T_t, so ACE quantifies the effect of changing image from $I_{i'}$ to I_i (with corresponding fusion knowledge change) while keeping trajectory fixed. This includes direct effects of I and its mediated effects through K, corresponding to the adjusted graph in Fig. 2(b).

C. Identifiability and Confounding

The identifiability of path-specific effects (TE, NDE_T, ACE) relies on the assumption that all relevant confounders are observed and the causal graph is correctly specified [28], [29]. In practice, however, ultrasound tracking involves latent confounders U that can bias estimation. We consider three key scenarios:

1) Visual Confounding ($U \rightarrow I \rightarrow P$): Visual confounding arises when unobserved factors, such as scanner characteristics or patient-specific anatomy, influence predictions through image features I. Since I is directly observed and its variation is implicitly modeled during training, this type of confounding is not the focus of our work.

2) Trajectory Confounding ($U \rightarrow T \rightarrow P$): Trajectory confounding occurs when latent variables induce biased trajectory patterns T, typically manifesting as periodicity bias where models overfit to repetitive motion cues. We mitigate this by removing the direct path from T to P.

3) Joint Confounding ($I \leftarrow U \rightarrow T$): Joint confounding emerges when a shared latent variable U simultaneously affects both I and T, introducing spurious correlations in the fused representation $K = f(I, T)$. This violates identifiability assumptions and calls for additional counterfactual deconfounding beyond what is addressed by the causal structure, in order to ensure reliable estimation of path-specific effects.

IV. Methodology

A. Overview

1) Causal Motivation: Conventional trackers suffer from two major issues: 1) *Periodicity bias* — over-reliance on trajectory priors creates a shortcut path $T \rightarrow P$, leading to underutilization of visual cues. This is particularly problematic in periodic motion, where spurious correlations between respiratory phases and target positions may dominate. 2) *Identifiability violation* — latent confounders U simultaneously affect both I and T (i.e., $I \leftarrow U \rightarrow T$), contaminating the fusion representation K and compromising the identifiability of path-specific causal effects.

2) Causal Solution Strategy and Architecture: We address these challenges through a dual-pronged causal strategy:

- **Path intervention:** We distinguish the beneficial mediated path ($T \rightarrow K \rightarrow P$) from the harmful direct shortcut ($T \rightarrow P$), eliminating the latter while preserving the former. This is achieved using two branches: the Total Effect (TE) Network estimates the overall influence of I and T on P, while the NDE_T Network blocks visual and fusion inputs to isolate the direct effect of T, which reflects the periodicity bias. Their difference yields a debiased prediction.
- **Deconfounding:** To address latent confounders, we incorporate a dedicated deconfounding module that removes spurious influence from the fused representation K, thereby restoring the assumptions required for valid path-specific effect estimation.

The following subsections detail the implementation of each component and the corresponding training procedure.

B. Structural Bias Removal via Path Intervention

To explicitly separate harmful periodicity bias from useful location priors, we design a path intervention mechanism using two dedicated networks: the Total Effect (TE) Network and the Natural Direct Effect of Trajectory (NDE_T) Network. These branches correspond to the full causal graph and trajectory-only subgraph, respectively, with their difference yielding the Adjusted Causal Effect (ACE) that eliminates the spurious direct path $T \rightarrow P$ while preserving the valid indirect path $T \rightarrow K \rightarrow P$.

1) Input Representation: Our framework processes three types of inputs to enable comprehensive causal modeling:

a) Visual Inputs: Template z and search image x are divided into patches, flattened, and projected into a latent space. Learnable positional embeddings are added to construct visual token sequences: $\mathrm{H}_v^0 = \left[\mathrm{H}_z^0;\ \mathrm{H}_x^0\right]$.

b) Trajectory inputs: To maintain token-level compatibility while modeling motion priors, we construct trajectory tokens from spatial, temporal, and dynamic characteristics. For each historical frame t, we extract center coordinates (x_t, y_t) and compute frame-rate-normalized displacement:

$$\Delta d_t = \frac{\|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2}{\delta t_t}, \tag{4}$$

where δt_t is the time interval between frames. Including temporal ordering through positional encoding $\mathrm{PE}(t)$, we form:

$$\mathbf{v}_t = \left[x_t,\ y_t,\ \Delta d_t,\ \delta t_t,\ \mathrm{PE}(t)\right]. \tag{5}$$

Following autoregressive tokenization strategies [14], [18], continuous components are discretized into finite vocabularies and mapped to learnable embeddings, yielding trajectory token sequence $\mathbf{H}_t = [\mathbf{h}_{t-T}^{\mathrm{traj}}, \ldots, \mathbf{h}_{t-1}^{\mathrm{traj}}]$.

c) Fusion knowledge inputs: Fusion knowledge K captures cross-modal dependencies through cross-attention within the Transformer encoder. Trajectory tokens attend to spatial patterns while visual tokens integrate motion priors. The resulting fused representation $K = f(I, T)$ serves as a causal mediator for indirect paths $I \rightarrow K \rightarrow P$ and $T \rightarrow K \rightarrow P$, enabling structural effect decomposition.

2) TE Network-Factual Prediction Branch: The TE Network estimates the total effect of template-search image I and historical trajectory T on target position P, covering all causal pathways. It employs a two-stream transformer encoder with three specialized branches:

Single-modal branches capture direct effects ($I \rightarrow P$ and $T \rightarrow P$), while the multi-modal branch models indirect effects mediated through $K = f(I, T)$. Fusion occurs at the feature level by jointly attending over final-layer tokens H_v^N and H_T^N from visual and trajectory encoders. Each branch uses dedicated prediction heads to generate similarity maps $\mathcal{A}_i$, $\mathcal{A}_t$, and $\mathcal{A}_k$ for visual, trajectory, and fused modalities. Each head is implemented using stacked Conv-BN-ReLU layers, which also decode offset $\mathcal{O}$ and normalized bounding box size $\mathcal{S}$ alongside the classification logits. The total-effect classification logit is:

$$\mathcal{P}_{i,t,k} = \mathcal{A}_i + \mathcal{A}_t + \mathcal{A}_k \tag{6}$$

To preserve semantic independence, we block gradient propagation across branches during backpropagation, ensuring each branch learns distinct representations aligned with its causal role. Visual encoding uses ViT backbone [30] for spatial-semantic modeling. Early candidate elimination [13] retains top k similarity candidates as foreground, focusing attention on discriminative regions.

3) NDE_T Network-Counterfactual Prediction Branch: The NDE_T Network estimates the direct effect of trajectory T on position P by simulating a counterfactual scenario where visual input I and multi-modal representation K are blocked.

a) Counterfactual simulation: Since neural networks require concrete inputs, we approximate blocked modalities using learnable surrogate distributions. Visual and fusion similarity maps are modeled as mixtures of m Gaussian distributions:

$$\mathcal{A}_{i'} = \sum_{c=1}^{m} \mathcal{N}(\mu_{ci}^{I}, \Omega_{ci}^{I}) \tag{7}$$

$$\mathcal{A}_{k'} = \sum_{c=1}^{m} \mathcal{N}(\mu_{ck}^{T}, \Omega_{ck}^{T}) \tag{8}$$

where μ and Ω denote mean and diagonal variance of each component. These mixtures serve as globally learnable surrogates representing the expected feature responses when I and K are blocked. They are optimized jointly with the network to provide differentiable counterfactual inputs, thereby enabling causal intervention in a stable and learnable form rather than using fixed population priors.

b) Bias estimation: The counterfactual logit map combines trajectory output with surrogate approximations:

$$\mathcal{P}_{t,i',k'} = \mathcal{A}_t + \mathcal{A}_{i'} + \mathcal{A}_{k'} \tag{9}$$

This reflects trajectory-induced effects absent visual information, interpreted as periodicity bias from the direct path $T \to P$.

C. Deconfounding for Identifiability Correction

While TE and NDE$_T$ networks enable path-specific effect estimation through structural intervention, their correctness depends on the identifiability of the underlying causal graph [29], [31]. In practice, unobserved variables U, such as ultrasound probe motion, patient respiration, or device-dependent gain settings [32], may simultaneously affect both trajectory T and visual input I (e.g., $I \leftarrow U \to T \to K \to P$), causing the fused representation $K = f(I, T)$ to encode confounding bias. This violates the Markovian assumption and contaminates the mediation pathway $K \to P$, undermining the identifiability of causal effects [33].

1) Proxy-Based Deconfounding Strategy: Although latent confounder U is unobservable, it leaves statistical traces in I and T through its influence on visual appearance and motion patterns. Inspired by proxy-based deconfounding [34], we approximate U using a latent variable $Z = g(I, T)$ learned from observed inputs. Rather than explicitly modeling Z in the causal graph [35], we adopt a removal strategy that eliminates Z's influence from K to prevent learning biased structural associations, considering the complexity and real-time requirements of visual tracking.

2) Deconfounding Architecture: Our deconfounding module employs a two-stage pipeline to extract and remove confounding influences:

- **Stage 1: Confounder Representation.** A variational encoder $g(I, T)$ infers latent confounder $Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$ under an information bottleneck constraint, encouraging Z to capture minimal confounding signals from the inputs.
- **Stage 2: Confounder Removal.** A decoder $F_{\text{deconf}}(K, Z)$ removes Z's influence from K, producing $\tilde{K}$. An adversarial discriminator guides this process by encouraging $\tilde{K}$ to be indistinguishable from K yet free of confounding bias.

3) Regularization Loss: We regularize the deconfounding process using an information bottleneck and an adversarial loss to ensure effective and minimal removal of latent confounders.

a) Information bottleneck regularization: To ensure that the latent variable Z effectively captures confounding signals shared between I and T, we employ an information bottleneck constraint inspired by variational information bottleneck principles [36], [37]:

$$\mathcal{L}_{\text{IB}} = \text{KL}(q(Z)\|p(Z)) + \lambda_{\text{var}} \cdot \text{Var}(\tilde{K}) \tag{10}$$

$$\text{Var}(\tilde{K}) = \frac{1}{B-1} \sum_{b=1}^{B} \|\tilde{K}_b - \bar{K}\|_2^2, \quad \bar{K} = \frac{1}{B} \sum_{b=1}^{B} \tilde{K}_b. \tag{11}$$

where $q(Z) = \mathcal{N}(\mu_Z, \sigma_Z^2)$ is the learned confounder distribution and $p(Z) = \mathcal{N}(0, I)$ is a standard Gaussian prior. The KL divergence term regularizes the confounder representation complexity, ensuring Z captures meaningful confounding patterns rather than noise. The across-sample variance penalty $\text{Var}(\tilde{K})$ measures the compactness of the deconfounded representation and indirectly reflects confounder-removal quality. When F_{deconf} effectively removes Z-related variations, $\tilde{K}$ becomes more concentrated across samples, indicating stable causal features.

b) Adversarial disentanglement: To effectively remove confounding influences from the fused representation, we introduce a Confounder Discriminator $D(\tilde{K})$ that distinguishes between original fused representations K (potentially confounded) and deconfounded representations $\tilde{K}$. The Deconfounding Decoder is trained adversarially to fool the discriminator through a minimax objective [38]:

$$\min_{F_{\text{deconf}}} \max_{D} \mathcal{L}_{\text{adv}} = \mathbb{E}_{K,Z}\big[\log D(K, Z) + \log(1 - D(\tilde{K}, Z))\big] \tag{12}$$

This adversarial process drives the decoder to remove all Z-predictable components from K, ensuring that $\tilde{K}$ and Z become statistically independent. As a result, $\tilde{K}$ retains only task-relevant causal information while removing non-causal variations induced by latent confounders.

4) Integration With the Causal Framework: The deconfounding module outputs a clean representation $\tilde{K}$ by removing confounder influence from the fused features K. This deconfounded representation is then used throughout the causal inference pipeline to ensure valid path-specific effect estimation.

Specifically, we replace K with $\tilde{K}$ in both the TE and NDE$_T$ branches, updating the predictive logits as follows:

$$\mathcal{P}_{i,t,\tilde{k}} = \mathcal{A}_i + \mathcal{A}_t + \mathcal{A}_{\tilde{k}} \tag{13}$$

$$\mathcal{P}_{t,i',\tilde{k}'} = \mathcal{A}_t + \mathcal{A}_{i'} + \mathcal{A}_{\tilde{k}'} \tag{14}$$

As illustrated in Fig. 3 [3], this adjustment structurally corrects the influence of latent confounders in the causal graph, restoring identifiability and enabling valid estimation of path-specific causal effects. In practice, the deconfounder is jointly trained with TE and NDE$_T$ networks, co-optimizing the information bottleneck and adversarial losses with the tracking objective to suppress confounding features.

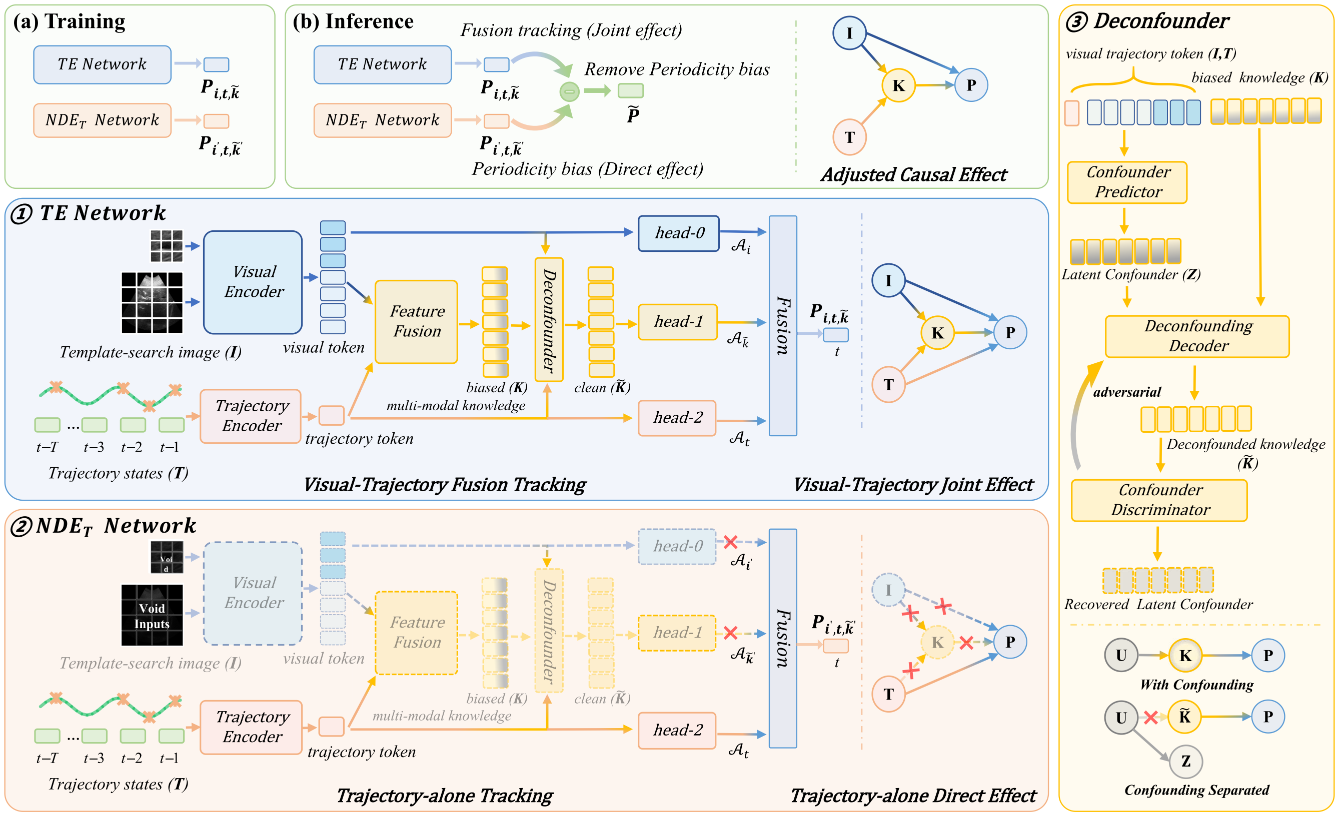


Fig. 3. **Network architecture and causal workflow. (a) Training stage [1] TE Network**, which models the total causal effect of image–trajectory pairs (I, T) on the predicted position P via both the direct path (T → P) and the indirect path through fused representation (I, T → K → P); **[2] NDE$_T$ Network**, which isolates the direct effect of trajectory T by blocking image input I and fused representation K, thereby capturing the periodicity bias introduced by T → P; **[3] Deconfounder Module**, which extracts latent confounders Z from (I, T) and removes their influence from K to obtain a deconfounded representation $\tilde{K}$, by separating proxy confounding signals to restore identifiability. **(b) Inference stage.** Final debiased prediction is computed by subtracting the direct trajectory-induced bias (estimated by NDE$_T$ Network) from the total effect (estimated by TE Network), enabling robust inference free from spurious correlations.

5) Residual Bias Suppression via Deconfounding: Although the Natural Direct Effect (NDE$_T$) removes the direct trajectory-to-prediction pathway ($T \to P$), residual periodic components may still leak into the mediated path ($T \to K \to P$). To further suppress such residual influence, the fused representation K is replaced with its deconfounded counterpart $\tilde{K}$.

This adjustment ensures that the adjusted causal effect (ACE) is computed as

$$\mathrm{ACE} = E[P(I_i, T_t, \tilde{K})] - E[P(I_{i'}, T_t, \tilde{K}')] \tag{15}$$

where $\tilde{K}$ and $\tilde{K}'$ denote the deconfounded fusion representations under different input configurations. Through a variational and adversarial process, the decoder $F_{\mathrm{deconf}}(K, Z)$ subtracts Z-predictable components from K. This not only eliminates shared confounding factors but also effectively attenuates residual periodic influence along $T \to K \to P$.

D. Loss Function

We define the overall training objective as a combination of tracking loss, branch consistency loss, and deconfounding regularization:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{track}} + \mathcal{L}_{\mathrm{KL}} + \mathcal{L}_{\mathrm{deconf}} \tag{16}$$

1) Tracking Loss: For the TE branch, we apply weighted focal loss $\mathcal{L}_{\mathrm{focal}}$ [39] for target classification and combine generalized IoU loss $\mathcal{L}_{\mathrm{IoU}}$ [40] with $\mathcal{L}_1$ loss for bounding box regression. The total tracking loss is:

$$\mathcal{L}_{\mathrm{track}} = \mathcal{L}_{\mathrm{focal}} + \lambda_1 \mathcal{L}_{\mathrm{IoU}} + \lambda_2 \mathcal{L}_1 \tag{17}$$

where λ_1 and λ_2 are empirically set, and for the trajectory-only NDE$_T$ branch, they are zero since bounding box supervision is unavailable.

2) Branch Consistency Regularization: To ensure the learnable surrogate distributions in NDE$_T$ yield meaningful counterfactual predictions, we introduce a KL divergence loss that regularizes the surrogate output $\mathcal{P}_{t,i',\tilde{k}'}$ toward the TE output $\mathcal{P}_{i,t,\tilde{k}}$:

$$\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}\left(\mathcal{P}_{t,i',\tilde{k}'} \| \mathcal{P}_{i,t,\tilde{k}}\right) \tag{18}$$

Here, the predictions are represented by logit maps parameterized as learnable Gaussian mixtures (see Eq. 8).

3) Deconfounding Regularization: As detailed in Section IV-C, we use a variational information bottleneck and adversarial disentanglement to remove latent confounder signals from the fused representation K:

$$\mathcal{L}_{\mathrm{deconf}} = \mathcal{L}_{\mathrm{IB}} + \lambda_{\mathrm{adv}} \cdot \mathcal{L}_{\mathrm{adv}} \tag{19}$$

E. Training and Inference

1) Training Strategy: Following standard protocols from [13], [14], we use CLUST 2D and NDTH datasets with a 4:1 random split. For CLUST dataset lacking bounding box annotations, we use fixed 32×32 square regions centered on annotated landmarks. The model is implemented in PyTorch 2.1 and trained for 100 epochs using two NVIDIA RTX 4090 GPUs. We adopt AdamW optimizer with learning rate 5×10^{-4} and batch size 32. Data augmentation includes random spatial translations and resizing, as well as random frame-rate sampling to enhance robustness against temporal resolution variations. We sample 500 ultrasound subsequences of length 32 per epoch, using temporal window $T = 16$ for trajectory representation.

2) Causal Inference Pipeline: During inference, we estimate debiased target locations by removing direct trajectory influence (periodicity bias) while retaining indirect contributions through fusion representation. The adjusted causal effect (ACE) is computed by subtracting counterfactual predictions from total effects:

$$\begin{aligned}\tilde{\mathcal{A}} &= \mathrm{ACE} \\ &= \mathbb{E}[P(I, T, \tilde{K})] - \mathbb{E}[P(I', T, \tilde{K}')] \\ &\approx \mathcal{P}_{i,t,\tilde{k}} - \mathcal{P}_{t,i',\tilde{k}'} \end{aligned} \tag{20}$$

Sigmoid activation produces the final classification probability map:

$$\tilde{\mathcal{P}} = \sigma(\tilde{\mathcal{A}}) \tag{21}$$

The peak value in $\tilde{\mathcal{P}}$ determines the predicted target location, refined using offset regression. Bounding box size is decoded directly from the TE Network, ensuring only bias-free, causally valid signals contribute to final predictions.

a) Autoregressive trajectory construction: Trajectory input T is built autoregressively from previous predictions. At each timestep t, a sliding window of length T collects historical estimates including predicted positions $\hat{\mathbf{p}}_{t'}$, displacements $\Delta\hat{d}_{t'}$, and time intervals $\delta t_{t'}$ for $t' = t-T, \ldots, t-1$. These features are tokenized as described in Section III-C3, with no ground-truth trajectory used at test time.

b) Cold-start handling: For initial frames ($t < T$), we warm-start the trajectory window using predictions from the TE Network's visual branch. Initial positions and pseudo-displacements are derived from similarity maps to form the first trajectory tokens, avoiding unrealistic assumptions like static targets or zero velocity.

V. Experiments

A. Datasets and Metrics

We evaluate our method on two liver ultrasound datasets: CLUST 2D (public) and NDTH (private).

1) CLUST 2D Dataset: The CLUST 2D dataset focuses on anatomical landmark tracking under respiratory motion [41][42]. It includes multi-center 2D ultrasound sequences from four subsets (CIL, ETH, ICR, MED) with 4–10 minute durations. We use 24 annotated sequences containing 53 anatomical landmarks with point coordinate annotations from three independent observers.

a) Evaluation metrics: Since CLUST provides only center point annotations, we employ center-based metrics: **ACE** (Average Center Error) measures pixel-wise Euclidean distance between predicted and ground-truth locations; $\mathbf{SR_3/SR_5}$ (Success Rate @ 3/5 pixels) represents the percentage of frames with prediction errors below the specified thresholds; $\mathbf{AUC_c}$ (Center-based Precision AUC) computes the area under the success rate curve across different center error thresholds, providing an overall precision measure.

2) NDTH Dataset: The NDTH dataset comprises real diagnostic ultrasound sequences acquired at the *Affiliated Drum Tower Hospital, Medical School of Nanjing University* using a LOGIQ E9 system (GE Healthcare, Chicago, IL, USA). It includes 129 patient sequences and 151 tracked targets. Three experienced radiologists selected subsequences with clearly visible landmarks and provided temporally consistent axis-aligned bounding boxes. Unlike the volunteer-based CLUST-2D dataset, NDTH contains real clinical cases involving vascular and tumoral structures, organized into four groups by spatial resolution and acquisition settings to support evaluation of model generalization.

a) Evaluation metrics: With bounding box annotations available, we employ both center-based and overlap-based metrics: **ACE** and $\mathbf{SR_5}$ maintain consistency with CLUST evaluation; **AO** (Average Overlap) calculates mean IoU between predicted and ground-truth bounding boxes; $\mathbf{AUC_i}$ (IoU-based Success AUC) measures the area under the success rate curve across different IoU thresholds, aligning with standard tracking protocols (Got-10K [43], LaSOT [44]).

B. Quantitative Comparison

We compare CUSTrack against state-of-the-art tracking methods across three categories: **Siamese-based** trackers (SiamFC [3], SiamDW [45], Ocean [46]); **Transformer-based** trackers (TransT [12], Stark [47], OSTrack [13], SeqTrack [14]); and **Temporal-aware** trackers (ARTrack [18], EVPTrack [19], AQATrack [20]). All methods were trained under the same experimental configuration, including data partition, random frame-rate sampling, and optimization settings, to ensure a fair comparison.

1) Performance on CLUST 2D: Table II shows that CUSTrack achieves superior performance with ACE of 2.99 and SR_5 of 97.13%. The generally lower ACE values on CLUST compared to NDTH reflect the dataset characteristics of small anatomical landmarks with point annotations. Siamese-based methods show limited performance, with SiamFC achieving ACE of 29.97 and Ocean improving to 20.44. Transformer-based trackers demonstrate substantial improvements, with OSTrack reaching ACE of 5.89 and SR_5 of 89.71% through effective feature interaction modeling.

Interestingly, ARTrack shows poor performance on CLUST (ACE of 13.78), likely due to two factors: the variable frame rates (6-31 Hz) across sequences disrupting autoregressive coordinate decoding, and the lack of bounding box annotations forcing the model to predict arbitrary box sizes from center points only. This overfitting to invalid target dimensions hampers the autoregressive approach. In contrast, CUSTrack's

TABLE II
QUANTITATIVE COMPARISON OF VARIOUS METHODS ON CLUST-2D, NDTH, AND MOTION-SPECIFIC SUBSETS. BOLD INDICATES THE BEST RESULTS AND UNDERLINED INDICATES THE SECOND-BEST RESULTS. * INDICATES WILCOXON SIGNED-RANK TEST $p < 0.05$, ** INDICATES $p < 0.01$(COMPARING THE BEST METHOD VS. SECOND BEST)

Method	CLUST-2D				NDTH				Periodic		Non-Periodic	
	ACE↓	SR_3(%)↑	SR_5(%)↑	AUC_c(%)↑	ACE↓	SR_5(%)↑	AO(%)↑	AUC_i(%)↑	ACE↓	SR_5(%)↑	ACE↓	SR_5(%)↑
SiamFC (2016)	29.97	55.80	66.93	68.33	26.18	70.99	69.42	68.14	27.31	68.82	32.18	67.08
SiamDW (2018)	28.95	51.50	60.14	65.44	9.75	65.72	66.67	65.70	16.05	56.44	18.17	75.97
Ocean (2020)	20.44	57.09	73.99	73.12	8.21	66.18	71.40	70.21	10.97	65.61	18.47	69.63
TransT (2021)	10.56	55.30	75.36	82.33	6.20	68.31	73.41	72.21	8.57	69.08	5.95	52.52
Stark (2021)	8.28	77.97	88.79	87.89	4.30	80.00	76.84	75.48	4.04	82.64	11.49	80.44
OSTrack (2022)	5.89	71.73	89.71	89.04	5.02	72.55	72.70	71.51	5.29	73.18	6.90	83.98
SeqTrack (2023)	4.58	59.47	86.87	91.17	5.38	57.11	72.11	70.95	5.46	62.61	4.68	76.92
ARTrack (2023)	13.78	29.17	61.83	78.51	4.55	72.72	74.15	72.90	6.21	74.56	15.44	61.68
EVPTrack (2024)	6.64	81.46	95.76	91.63	4.49	79.39	73.90	72.64	**3.30**	**82.29**	8.93	89.75
AQATrack (2024)	3.99	68.88	94.50	91.94	3.62	76.98	76.37	75.12	3.51	79.94	6.80	89.58
CUSTrack (Ours)	**2.99****	**88.89****	**97.13**	**93.62***	**3.40***	**80.88**	**77.51***	**76.10**	3.31	82.19	**3.90****	**92.24***

significant improvement demonstrates the effectiveness of causal debiasing in addressing trajectory-induced periodicity bias while maintaining robustness to annotation inconsistencies.

2) Performance on NDTH: NDTH presents a more challenging scenario with consistently higher ACE values across all methods, reflecting clinical variations, diverse imaging conditions, and larger target structures. CUSTrack maintains superiority with ACE of 3.40 and SR_5 of 80.88%. Notably, ARTrack shows improved performance on NDTH (ACE of 4.55) compared to its poor CLUST results, validating that consistent frame rates and proper bounding box annotations enable effective autoregressive decoding. However, some temporal-aware methods still struggle with irregular clinical motion patterns, emphasizing the robustness of our causal framework in handling complex motion scenarios.

3) Motion-Specific Subset Analysis: To assess performance under different motion regimes, we derive two diagnostic subsets based on frequency-domain analysis of landmark trajectories: **periodic sequences** (28) dominated by cyclic respiratory patterns, and **non-periodic sequences** (7) with irregular motion. These evaluation-only subsets enable targeted assessment of causal intervention effectiveness using ACE and SR_5 metrics across motion types.

The motion-specific evaluation provides crucial insights into our causal framework's effectiveness. On periodic sequences, CUSTrack achieves competitive performance (ACE of 3.31, SR_5 of 82.19%), demonstrating that our method preserves beneficial motion priors inherent in cyclic patterns. EVPTrack performs slightly better on periodic data (ACE of 3.30), confirming that periodic motion naturally aligns with trajectory-based approaches.

However, the critical advantage emerges on non-periodic sequences, where CUSTrack significantly outperforms all competitors with ACE of 3.90 and SR_5 of 92.24%. Methods heavily relying on trajectory priors show substantial performance degradation: ARTrack's ACE deteriorates from 6.21 to 15.44, demonstrating how autoregressive approaches struggle when motion patterns become unpredictable. This pattern reveals that conventional temporal methods fail when historical motion patterns become unreliable.

TABLE III
TRACKING PERFORMANCE COMPARISON UNDER DIFFERENT CAUSAL INFERENCE STRUCTURES

Method	CLUST-2D			NDTH		
	ACE↓	SR_3(%)↑	SR_5(%)↑	ACE↓	SR_5(%)↑	AO(%)↑
V1	7.38	63.25	91.90	4.59	64.50	74.46
V2	4.55	83.23	94.39	4.53	77.09	75.57
V3	3.78	86.66	96.02	4.09	73.94	76.14
CUSTrack (Ours)	**2.99**	**88.89**	**97.13**	**3.40**	**80.88**	**77.51**
V4	10.33	59.13	84.07	6.78	63.43	61.28
V5	10.18	57.54	86.00	7.86	58.77	69.12

Our results validate the core hypothesis: while trajectory information contains valuable spatial constraints (beneficial in periodic scenarios), it also introduces harmful periodicity bias that degrades performance when motion deviates from expected patterns. CUSTrack's causal framework successfully disentangles these effects, preserving helpful motion priors while eliminating detrimental biases, leading to robust performance across diverse motion regimes.

C. Effect of Causal Structures for Unbiased Tracking

We evaluate the impact of different causal inference structures using the variants illustrated in Fig. 4 and summarized in Table III. Variant V1 introduces trajectory priors only during training without causal modeling at inference. It shows the weakest performance, particularly on NDTH (SR_5: 64.50%). Variant V2 retains the visual branch during inference but omits causal disentanglement, yielding moderate gains (ACE: 4.55 on CLUST-2D). Variant V3 removes the direct image path entirely, relying solely on mediated effects, which leads to further improvements in CLUST-2D (ACE: 3.78) but reduced robustness on NDTH (AO: 76.14). Variants V4 and V5 perform comparably and remain inferior to the other designs across both datasets (ACE >10 on CLUST-2D), since periodicity bias is still captured either through the learnable trajectory weight in V4 or through direct mixing into the fused representation in V5. These results confirm that naive modulation

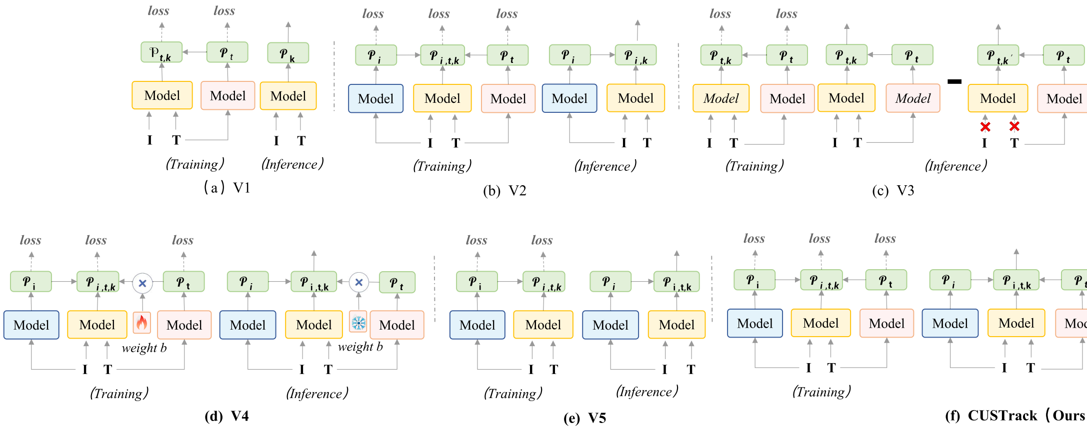


Fig. 4. Illustration of causal structures of CUSTrack and its variants. (a) **V1** that introduces an additional branch to capture trajectory priors during the training stage and removes it during inference; (b) **V2** differs from our CUSTrack during inference, which directly fuses the visual branch for prediction; (c) **V3** differs from our CUSTrack by removing the visual branch both during training and inference stages; (d) **V4** that assigns a learnable weight b to the trajectory branch and fuses $A_i + A_k + b \cdot A_t$ at inference; (e) **V5** that removes the direct $T \rightarrow P$ path by disabling the trajectory head, leaving trajectory cues only mixed into K. (f) **CUSTrack** makes debiased prediction via the removal of direct effect of trajectory states.

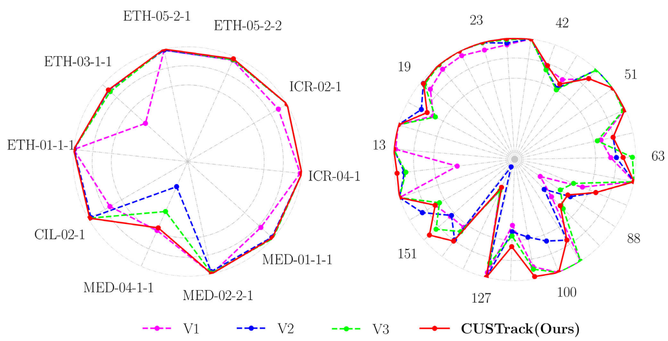


Fig. 5. Success rate (SR_5) comparison of CUSTrack variants across different sequences. NDTH results use size-normalized thresholds.

or shortcut removal without explicit deconfounding cannot effectively mitigate respiratory bias.

In contrast, CUSTrack achieves the best overall results across both datasets, with ACE of 2.99 on CLUST-2D and 3.40 on NDTH, validating the effectiveness of our full causal structure in mitigating trajectory bias and improving generalization. We also visualize SR_5 across sequences in Fig. 5, highlighting the consistent performance advantage of CUSTrack over its ablated variants.

D. Effect of Assumptions for Counterfactual Outputs

We investigate the impact of different assumptions for generating counterfactual outputs within our framework. As baselines, the "Random" assumption simulates uninformed guessing of the target location, while the "Prior" assumption uses the empirical mean distribution from the training set. Since B-mode ultrasound speckle statistics are often modeled by Rayleigh distributions, we further examine whether such a distributional assumption benefits counterfactual inference. To this end, we introduce a learnable Rayleigh surrogate (L-Rayleigh) that represents the counterfactual similarity maps as mixtures of several Rayleigh components with learnable centers and scales.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT ASSUMPTIONS FOR COUNTERFACTUAL OUTPUTS. L-RAYLEIGH AND L-GAUSSIAN INDICATE LEARNABLE RAYLEIGH AND LEARNABLE GAUSSIAN MODELING, RESPECTIVELY

Method	CLUST-2D			NDTH		
	ACE↓	SR_3(%)↑	SR_5(%)↑	ACE↓	SR_5(%)↑	AO(%)↑
Random	9.51	68.10	86.81	8.83	71.61	71.20
Prior	5.00	78.39	95.80	4.56	71.34	75.34
L-Rayleigh	4.57	82.90	96.85	4.77	74.99	76.17
L-Gaussian (Ours)	**2.99**	**88.89**	**97.13**	**3.40**	**80.88**	**77.51**

As shown in Table IV, the Random strategy performs poorly, with ACE of 9.51 on CLUST-2D and 8.83 on NDTH. Prior improves overall performance, but still lags behind our learnable approach. Compared with Prior, L-Rayleigh yields consistent improvements, reducing ACE to 4.57 on CLUST-2D and 4.77 on NDTH. However, its performance remains inferior to our Gaussian surrogate. Specifically, our method achieves the lowest ACE (2.99 and 3.40) and highest SR_5 and AO values on both datasets. These results demonstrate that learning a distribution for counterfactual inference is crucial for improving robustness in ultrasound tracking, indicating that while Rayleigh modeling captures the underlying speckle statistics, Gaussian modeling not only offers mathematical simplicity and symmetric target-centered kernels but also provides superior empirical robustness for ultrasound tracking.

E. Effect of Deconfounding Module

To assess the impact of the deconfounding module, we compare the tracking success rates of the full CUSTrack model and a variant without the deconfounding component (w/o Deconf) on different groups of sequences in the CLUST-2D dataset. As illustrated in Fig. 6, CUSTrack consistently outperforms the baseline across all sequence groups, with particularly notable improvements in sequences exhibiting complex motion or ambiguous appearance. These results indicate that the

TABLE V
COMPARISON OF TEMPORAL DECONFOUNDING STRATEGIES ON CLUST-2D AND NDTH. BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST RESULTS. ✓, ✘, AND ✗ REPRESENT FULL, NONE, AND PARTIAL PRESERVATION OR REMOVAL OF MOTION PRIORS AND PERIODIC BIAS

Method	Prior Preserved	Bias Removed	CLUST-2D				NDTH			
			ACE↓	SR_3(%)↑	SR_5(%)↑	AUC_c(%)↑	ACE↓	SR_5(%)↑	AO(%)↑	AUC_i(%)↑
Trajectory Reshuffling	✘	✓	7.49	78.32	95.38	90.48	4.93	73.78	74.01	72.78
Random Frame-rate Sampling	✓	✘	6.62	65.57	87.02	87.63	5.13	<u>77.26</u>	74.55	73.30
Temporal Dropout	✗	✗	5.94	80.39	92.44	90.76	<u>4.17</u>	72.92	<u>75.00</u>	<u>73.74</u>
Adversarial Deconfounding	✓	✗	<u>4.20</u>	<u>83.10</u>	<u>94.67</u>	<u>91.61</u>	4.41	70.80	73.27	72.05
CUSTrack (Ours)	✓	✓	**2.99**	**88.89**	**97.13**	**93.62**	**3.40**	**80.88**	**77.51**	**76.10**

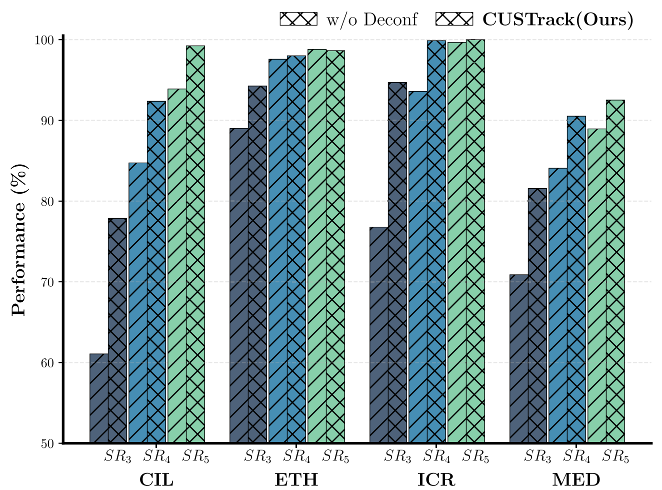


Fig. 6. Comparison of SR on grouped sequences in CLUST-2D with and without the deconfounding module.

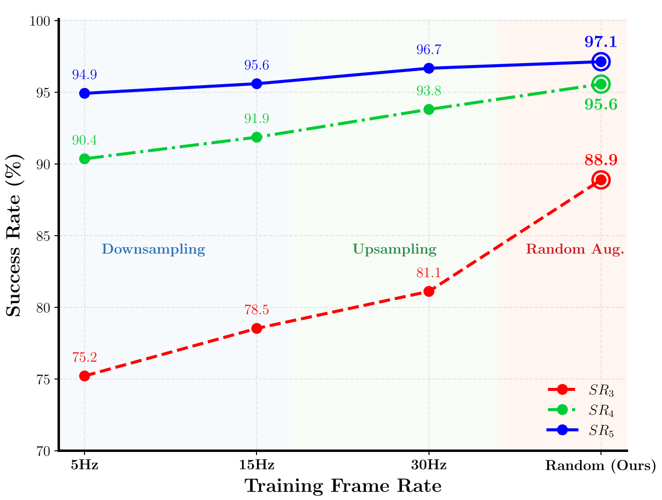


Fig. 7. Comparison of SR on CLUST-2D under different training frame rates (5 Hz, 15 Hz, 30 Hz, and random sampling (ours)).

deconfounding module effectively improves the identifiability of trajectory-related variables, enhancing robustness against spurious correlations and appearance-induced confounding.

F. Comparison of Frame Rate Sampling Strategies

We evaluate the effect of training frame rates on model performance. As shown in Fig. 7, models trained with fixed downsampling or upsampling (5 Hz, 15 Hz, and 30 Hz) exhibit gradually improved results as the temporal resolution increases, yet remain limited by their reliance on a single sampling frequency. In contrast, the proposed random frame-rate sampling achieves the highest overall success rates (SR_3, SR_4, and SR_5), consistently outperforming fixed-rate strategies. These results indicate that exposing the model to heterogeneous temporal intervals enhances generalization across motion speeds and acquisition settings, while preventing frame rate from acting as a confounding factor. Random frame-rate sampling therefore provides both empirical robustness and methodological consistency with realistic ultrasound scenarios, where frame rates typically vary between 6 Hz and 31 Hz.

G. Comparison With Temporal Deconfounding Strategies

To compare our temporal deconfounding strategy with other bias-mitigation approaches, we evaluate three representative categories of temporal deconfounding methods, as summarized in Table V.

1) Data-Level Augmentation: methods include *Trajectory Reshuffling* and *Random Frame-rate Sampling*. The former randomly permutes historical frames, eliminating the cyclic dependency between trajectory and prediction but also removing informative motion priors, resulting in the weakest accuracy (ACE: 7.49 on CLUST-2D). The latter samples at a random fixed rate for uniform temporal spacing, preserving rhythmic motion yet retaining periodic bias (ACE: 6.62).

2) Model-Based Regularization: methods, such as *Temporal Dropout* [48], introduce temporal regularization by selectively dropping salient temporal or spatial responses during training to reduce overfitting to repetitive motion patterns while preserving essential temporal dynamics. On CLUST-2D, it improves performance to ACE 5.94 and SR_3 80.39%, showing moderate alleviation of periodic bias.

3) Causal Deconfounding: approaches comprise *Adversarial Deconfounding* and our proposed *CUSTrack*. Adversarial Deconfounding adopts an adversarial feature alignment strategy to learn representations invariant to periodic or appearance-related bias [49], [50], preserving motion priors but only partially mitigating respiratory bias since it suppresses superficial confounding effects without explicitly modeling temporal causal dependencies. As shown in Table V, it achieves ACE of 4.20 and SR_3 83.10% on CLUST-2D, while its improvement on NDTH remains marginal (ACE 4.41).

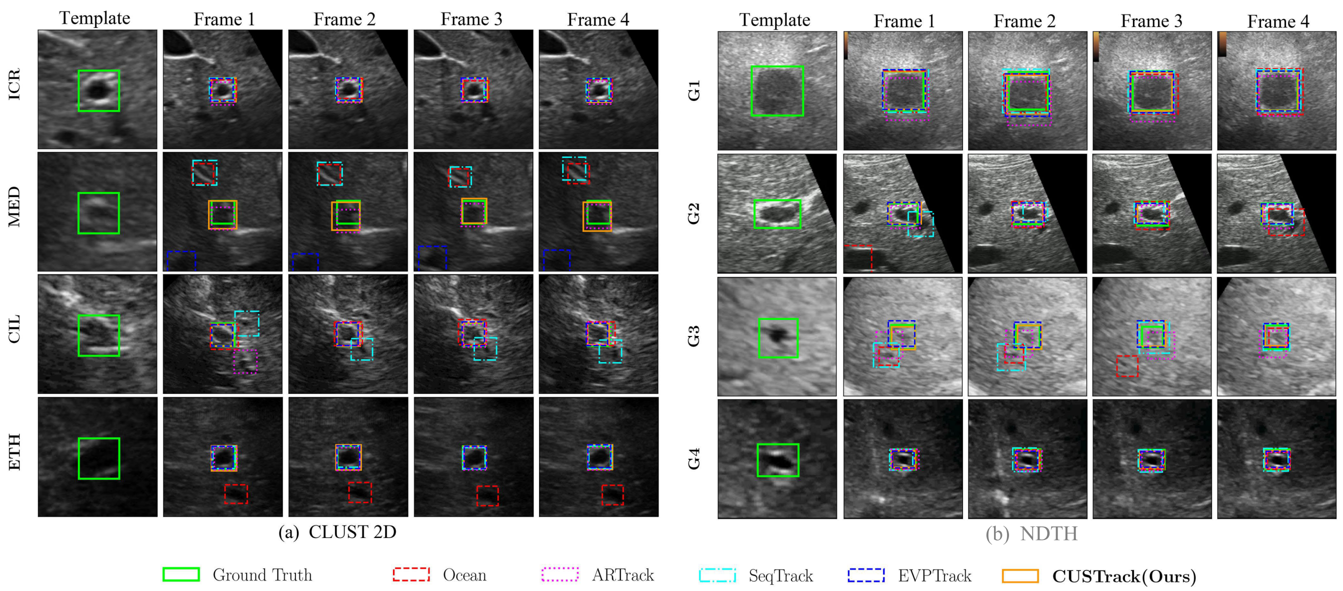


Fig. 8. Visualization of tracking results across various methods. Each row corresponds to one sample selected from different ultrasound sequences. The green bounding box outlines the ground truth, while the colored boxes show the tracking results of the respective methods.

In contrast, CUSTrack performs explicit path-specific intervention to remove the direct trajectory-to-prediction effect while retaining beneficial motion cues, achieving the best results across both datasets (ACE: 2.99 on CLUST-2D and 3.40 on NDTH). These comparisons demonstrate that while data-level perturbations and heuristic regularization offer limited bias alleviation, causal deconfounding—especially with explicit temporal intervention—provides a principled and more effective solution for robust ultrasound tracking under periodic motion.

H. Qualitative Evaluation

1) Spatial Tracking Accuracy: Fig. 8 presents qualitative comparisons of tracking results across diverse ultrasound sequences. For relatively stable scenarios, such as ICR, ETH, G1, and G4, most methods track the target accurately, with only minor drift observed in Ocean due to its convolution-based architecture, which struggles with subtle target deformations. In contrast, for more challenging sequences—such as MED, G2, and G3—where the target undergoes large deformations or becomes barely visible, visual-only trackers (e.g., Ocean and SeqTrack) occasionally lose the target entirely. Moreover, temporal-aware methods such as EVPTrack (on MED) and ARTrack (on CIL and G3) also show varying degrees of drift, likely due to misalignment between motion priors and actual dynamics. In all sequences, CUSTrack consistently produces precise localization and tightly aligned trajectories, demonstrating robust performance against both appearance degradation and motion uncertainty.

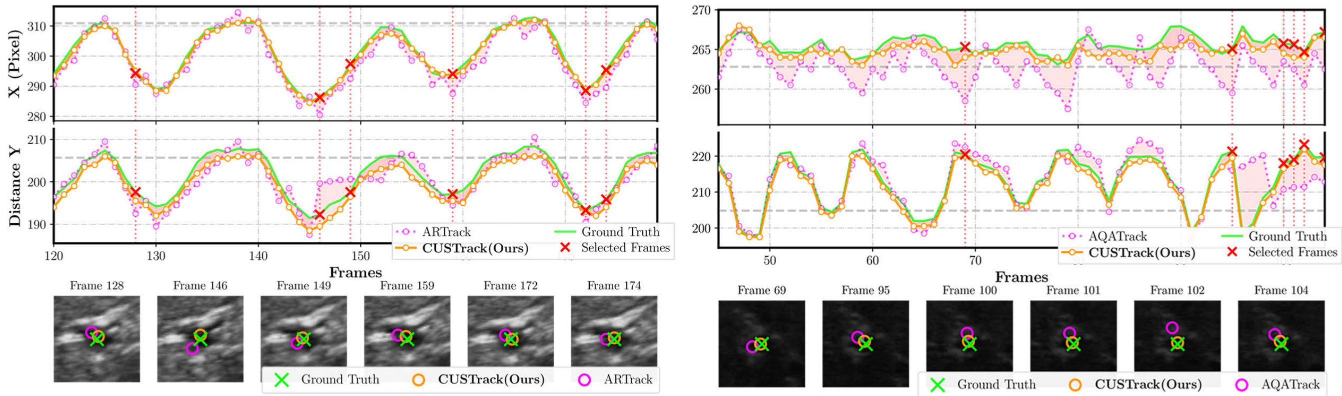
2) Temporal Tracking Behavior: As illustrated in Fig. 9, (a) compares the temporal tracking behaviors of CUSTrack with two temporal-aware methods, ARTrack and AQATrack. The plots show the variations of target center coordinates (X and Y), both exhibiting cyclic patterns consistent with respiratory motion. In the ARTrack comparison, the Y-coordinate shows a clear frequency shift around frame 146, during which ARTrack continues to predict following the original rhythm, leading to noticeable tracking errors. Similarly, AQATrack fails to adapt to changes in motion amplitude and frequency, resulting in cumulative drift during periodic transitions. In contrast, CUSTrack accurately follows the ground-truth trajectory in both subsequences, maintaining alignment even when the respiratory cycle changes.

In Fig. 9(b), we further compare CUSTrack with its internal sub-networks, TE Network and NDE$_T$ Network. The TE Network remains partially affected by periodicity bias, producing erroneous cyclic predictions between frames 83 and 100. In contrast, the NDE$_T$ Network behaves as an almost purely periodic predictor, generating repetitive trajectories that fail to capture non-periodic motion variations. CUSTrack effectively overcomes these limitations through its causal disentanglement mechanism, achieving stable and adaptive temporal modeling across both periodic and irregular respiratory patterns. These results confirm that CUSTrack's superiority extends beyond spatial accuracy to consistent and causally grounded temporal tracking.

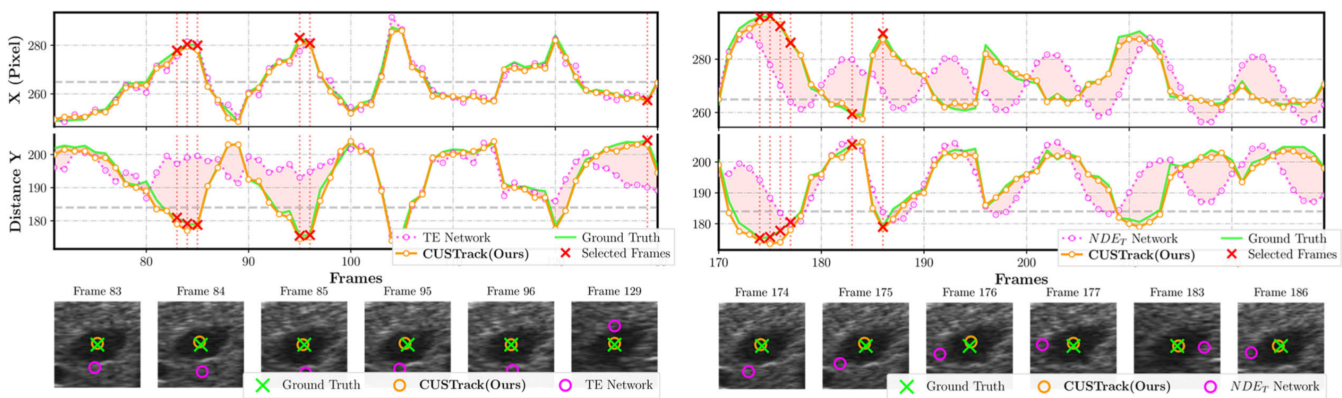
VI. Discussion

A. Runtime Performance Analysis

We evaluate the computational efficiency of CUSTrack and several representative trackers, as shown in Fig. 10. All methods are tested on an NVIDIA RTX 4090 GPU under identical settings (256 × 256 input, batch = 1). Our model achieves a favorable balance between accuracy and inference speed, satisfying the real-time requirement of ultrasound imaging [42]. The efficiency gains primarily originate from three architectural designs. First, a *candidate elimination mechanism*[13] is integrated into the visual backbone to progressively prune redundant visual tokens, thereby reducing the computation



(a) Temporal tracking visualization of ARTrack, AQATrack, and the proposed CUSTrack.



(b) Temporal tracking visualization of the TE Network and $\mathrm{NDE_T}$ Network within CUSTrack.

Fig. 9. Visualization of temporal tracking behaviors across different methods. Each block compares tracking trajectories on a randomly selected ultrasound sequence, where the green curve indicates the ground truth and the highlighted points correspond to representative frames shown below.

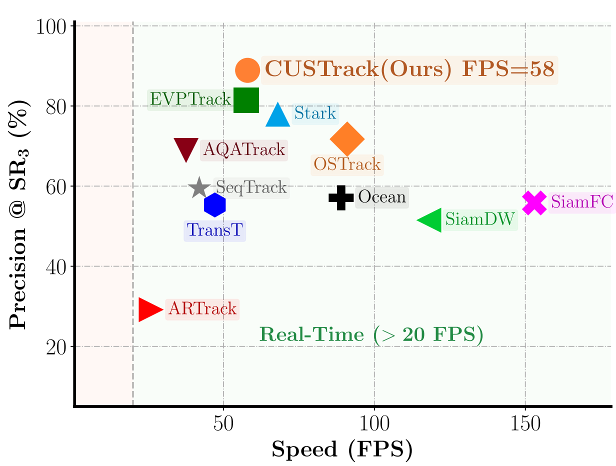
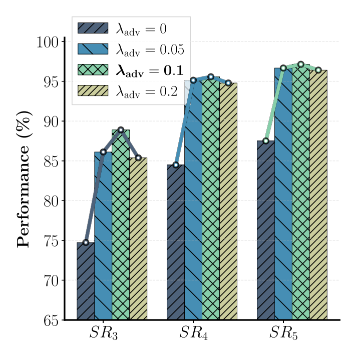
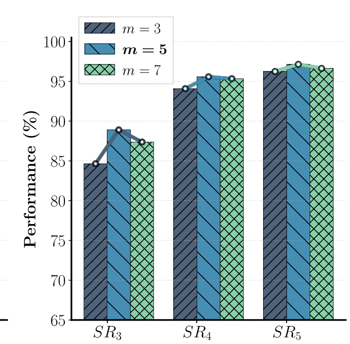


Fig. 10. Precision–speed comparison of CUSTrack and compelling tracking methods on CLUST-2D.



(a) λ_{adv} sensitivity.



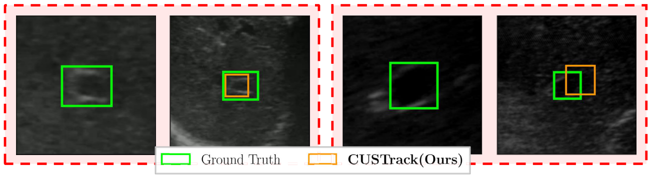
(b) m sensitivity.

Fig. 11. Sensitivity analysis of adversarial weight λ_{adv} and Gaussian component number m.

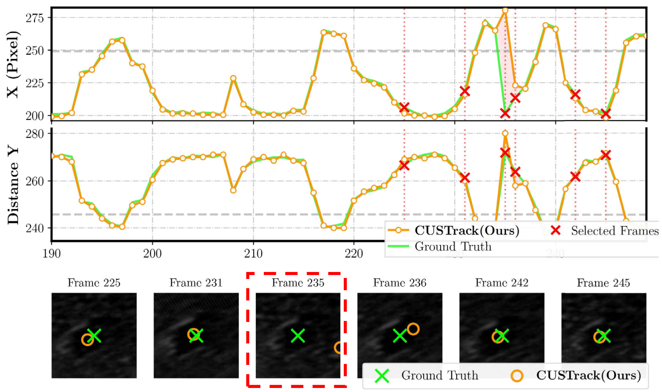
of subsequent Transformer layers. Second, in the trajectory prediction branch, *cached encoder features* are reused and combined with a lightweight coordinate encoding scheme to minimize redundant operations. Third, the *counterfactual reasoning* is implemented using a small number of learnable parameters rather than additional deep subnetworks, which avoids extra computational burden. These strategies collectively enable CUSTrack to maintain real-time performance while preserving high tracking precision, achieving an effective trade-off between accuracy and efficiency for practical ultrasound applications.

B. Hyperparameter Stability Analysis

We further examine the stability of key hyperparameters in CUSTrack, including λ_1 and λ_2 in Eq. 17, the adversarial weight λ_{adv} in Eq. 19, and the Gaussian component number m



(a) Tracking drifts caused by acoustic shadowing and blurred boundaries.



(b) Temporal tracking failure under irregular motion and low-visibility frames.

Fig. 12. Representative failure cases of CUSTrack on the CLUST-2D dataset from static and temporal perspectives.

in the counterfactual surrogate (Eq. 8). Following common practice in visual tracking [13], [47], λ_1 and λ_2 are fixed to 2.0 and 5.0. We then vary $\lambda_{\text{adv}} \in \{0, 0.05, 0.1, 0.2\}$ and $m \in \{3, 5, 7\}$ to assess robustness. As shown in Fig. 11, setting $\lambda_{\text{adv}} = 0$ disables the adversarial module and causes a clear accuracy drop, whereas for $\lambda_{\text{adv}} > 0$, SR_5 fluctuates within 2%, indicating low sensitivity and an optimum near 0.1. Similarly, $m = 5$ yields the best performance, with SR_3 and SR_5 variations below 5% and 3%, respectively. These results demonstrate that CUSTrack remains stable under moderate hyperparameter variations, reflecting the inherent robustness of its causal formulation.

C. Failure Cases Analysis

Although CUSTrack demonstrates stable tracking performance across most sequences, several representative failure cases are observed, as illustrated in Fig. 12. As shown in Fig. 12(a) and Fig. 12(b), both types of failures share a common cause: the target region becomes visually indistinct and lacks consistent motion patterns, making *both appearance features and historical motion information unreliable*. In Fig. 12(a), when acoustic shadowing or blurred anatomical boundaries occur, the tracker cannot extract sufficient visual cues, and the irregular temporal dynamics further prevent effective motion compensation, leading to spatial misalignment. Fig. 12(b) shows a temporal tracking sequence where the target region remains ambiguous for several frames, causing temporary localization failure; once the motion becomes regular again or the structure regains visual distinguishability, the causal memory in CUSTrack enables rapid recovery and suppresses long-term drift. These observations indicate that the model's causal design improves robustness to moderate noise and periodic bias but still struggles when both visual recognition and temporal regularity break down, inspiring our future efforts toward enhancing the network's visual discrimination capability under complex ultrasound conditions.

D. Clinical Relevance and Future Directions

While the proposed causal tracking model may not further reduce minimal frame-wise error on the public CLUST dataset, it provides substantial gains in tracking stability for ultrasound-guided interventions. By explicitly removing spurious periodic dependencies, CUSTrack maintains smoother and more reliable localization under varying respiratory or probe-induced motion. Such enhanced stability is clinically beneficial for radiotherapy dose delivery, biopsy guidance, and tumor monitoring, where consistent tracking is more critical than marginal per-frame accuracy improvements. These findings highlight the translational potential of causal modeling in ultrasound tracking and motivate future exploration of patient-specific adaptation and real-time clinical deployment.

VII. Conclusion

We revisit liver ultrasound tracking from a causal perspective and propose CUSTrack, a framework that mitigates periodicity bias arising from periodic motion. By identifying the direct effect of historical trajectories as a source of spurious correlations, we apply counterfactual reasoning to subtract biased influences and introduce a deconfounding module to ensure causal identifiability. Experiments on two datasets show that CUSTrack improves tracking accuracy and robustness under diverse motion patterns.

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